

Digital Forensics Investigation of Cloud Storage Artefacts on Windows 11

**A Practitioner Case Study of OneDrive, Google Drive and Encryption-Related
Artefacts**

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Abstract

Cloud storage creates an important challenge for digital forensic investigations because relevant information may be held on provider-controlled infrastructure while the Windows endpoint retains local traces of cloud-service activity. This practitioner case study presents the findings of a controlled forensic investigation of Microsoft OneDrive and Google Drive on a Windows 11 25H2 virtual-machine environment. Four scenarios were examined: active synchronisation, file deletion, account sign-out and a local encryption simulation. Evidence sources included Windows Registry artefacts, Prefetch, Windows Event Logs, cloud-client SQLite databases and volatile memory analysed with Volatility 3.

The investigation demonstrated that useful endpoint evidence remained recoverable across the tested scenarios, although persistence varied by artefact type and system state. OneDrive Registry evidence remained available after file deletion and account sign-out. Cloud-client database records also preserved deletion-related information in the tested deletion scenario, while local database contents were cleared following account sign-out. Google Drive database analysis identified retained metadata, including MD5-related information associated with deleted test files. In the encryption scenario, memory analysis identified VeraCrypt.exe and veracrypt.sys after the encrypted container had been dismantled. These findings support a multi-source investigative approach rather than reliance on a single cloud-client artefact.

The case study concludes by presenting how the empirical findings informed a structured thirteen-stage investigation framework covering preservation, acquisition, artefact analysis, cross-source correlation and reporting. The work is situated within a UK forensic context and considers the evidential principles addressed by ACPO guidance and the Police and Criminal Evidence Act 1984. The findings are limited to the controlled Windows 11 environment and should not be interpreted as universal artefact behaviour across all operating-system or cloud-client versions.

1. Introduction

Cloud services such as Microsoft OneDrive and Google Drive are widely used for storing, synchronising and accessing files. For forensic investigators, this creates a practical problem: cloud-hosted content may not be directly available from a seized endpoint, yet the local Windows system may retain artefacts showing that a cloud service was configured, executed or used.

Previous cloud-forensics research has demonstrated the forensic value of endpoint artefacts, including Registry information and local application databases. However, the project underlying this case study addressed a more specific combination of conditions: Windows 11 endpoint analysis, simultaneous OneDrive and Google Drive coverage, controlled deletion and sign-out scenarios, an encryption simulation, and consideration of UK evidential requirements.

The purpose of the investigation was therefore to determine which artefacts persisted under these conditions and to use the empirical observations to construct a practical investigation methodology.

2. Investigation Objectives

The investigation sought to identify OneDrive and Google Drive artefacts on Windows 11, examine persistence following file deletion and account sign-out, investigate volatile evidence of encryption software, correlate multiple evidence sources, develop a structured forensic framework, and consider preservation and evidential-handling requirements within a UK context.

3. Experimental Design

A controlled Windows 11 25H2 virtual-machine environment was used. Microsoft OneDrive and Google Drive for Desktop were installed and used with researcher-generated test data. Four scenarios were executed from a controlled baseline so that changes in endpoint artefacts could be compared between conditions.

Scenario 1 established the active-synchronisation baseline. Scenario 2 examined the endpoint after test files were deleted from local and cloud storage locations. Scenario 3 examined the endpoint after cloud accounts were disconnected. Scenario 4 examined a local encryption workflow in which an encrypted container was created, used and dismantled before forensic analysis.

Evidence was acquired and analysed using forensic tools including RECcmd, PECmd, EvtxECmd, Volatility 3, Magnet RAM Capture and DB Browser for SQLite. The investigation used targeted artefact acquisition rather than relying exclusively on a complete forensic disk image. The project records limitations affecting RAM acquisition in Scenario 1 and full disk imaging.

4. Artefact Sources and Analysis

Evidence source	Primary tool	Investigative purpose
Windows Registry	RECcmd	OneDrive account and application persistence
Prefetch	PECmd	Application execution evidence and timing
Windows Event Logs	EvtxECmd	System and temporal context
Cloud-client SQLite	DB Browser for SQLite	Synchronisation, file and metadata evidence
RAM	Volatility 3	Processes, modules and volatile application evidence

5.1 Registry Persistence

RECcmd analysis identified 25 OneDrive-related Registry key hits in Scenario 1 and 23 in Scenario 2. The location Software\Microsoft\OneDrive\Accounts\Personal remained available after the tested file-deletion procedure. Relevant OneDrive Registry information also remained after account sign-out. This demonstrates that cloud-client database absence cannot by itself establish that a cloud service had never been used on the endpoint.

5.2 Cloud-Client Database Evidence

OneDrive SyncEngineDatabase.db and Google Drive local databases provided file and synchronisation evidence. In the deletion scenario, OneDrive records retained deletion-related information and Google Drive mirror_sqlite retained MD5-related information for the tested files. Following account sign-out, the tested cloud-client databases were empty or contained no file-level records. The contrast illustrates that persistence is dependent on artefact type and endpoint state.

5.3 Memory Evidence of Encryption

The encryption scenario produced a significant volatile-memory finding. Volatility 3 identified VeraCrypt.exe (PID 12408) and the veracrypt.sys kernel driver in the acquired RAM. The encrypted container had already been dismantled before acquisition. Consequently, memory analysis provided endpoint evidence of encryption-software activity without requiring the encrypted container to remain mounted.

5.4 Cross-Source Correlation

The investigation showed the value of correlating evidence sources. Registry evidence provided persistence information; Prefetch supported execution analysis; SQLite databases provided cloud-client metadata; Event Logs provided system context; and RAM provided volatile process and module information. No individual artefact category was treated as sufficient on its own.

5. Key Findings

The principal empirical findings are presented below.

6. Practitioner Investigation Framework

The empirical findings informed a thirteen-stage investigation framework:

1. Pre-acquisition documentation
2. Live RAM capture
3. Registry export
4. Artefact export
5. System shutdown
6. Registry analysis
7. Prefetch analysis
8. Event Log analysis
9. OneDrive database analysis
10. Google Drive database analysis
11. Memory analysis
12. Cross-source correlation
13. Evidence reporting

The framework places preservation and acquisition before detailed examination and requires findings to be correlated across multiple evidence sources. It is intended for authorised forensic investigations and is not a method for unauthorised access to cloud services or devices.

7. UK Evidential Context

The original project considered the ACPO Good Practice Guide for Digital Evidence and the Police and Criminal Evidence Act 1984 when designing its preservation, acquisition and documentation approach. This case study does not provide legal advice or assert that the experimental results are automatically admissible in court. The UK context is used to emphasise preservation, documentation, integrity and lawful authority.

ACPO guidance is historical material; investigators should consult current applicable professional, organisational and legal guidance when conducting a real investigation.

8. Limitations

- The investigation used a controlled Windows 11 virtual-machine environment and researcher-generated test data.
- Only selected cloud platforms and endpoint artefact sources were examined.
- RAM acquisition for Scenario 1 was unsuccessful because of a disk-space interruption, so that memory image could not be parsed successfully.
- Full disk imaging could not be completed as originally planned because of VDI/FTK Imager and local storage constraints documented in the project.
- No external practitioner validation was performed.
- Artefact structures and persistence behaviour may vary with Windows, cloud-client and application versions.
- The findings should therefore be interpreted as controlled experimental observations rather than universal artefact-presence rules.

9. Practical Implications

The findings indicate that cloud-forensics examinations should not stop when cloud-hosted files are unavailable or when a local cloud-client database has been cleared. Registry artefacts, Prefetch, Event Logs, remaining database metadata and volatile memory may provide complementary evidence.

The investigation also demonstrates the importance of acquiring volatile evidence early where encryption or other transient application activity is relevant. The Scenario 4 result illustrates how RAM can preserve evidence of encryption activity after the associated encrypted container has been dismantled.

10. Conclusion

This controlled investigation demonstrated that Windows 11 endpoints can retain useful forensic traces of OneDrive and Google Drive activity under active synchronisation, file deletion, account sign-out and encryption conditions. The persistence and availability of evidence differed between artefact categories, supporting a multi-source forensic methodology.

The principal contribution of the project is the systematic combination of endpoint cloud artefact analysis, volatile-memory analysis and UK evidential considerations into a structured investigation framework. The framework translates the empirical observations into a practical sequence covering preservation, acquisition, analysis, correlation and reporting.

Further validation using additional Windows versions, cloud-client versions, cloud platforms, full-disk acquisitions and independent practitioner testing would strengthen the generalisability of the framework.

11. Selected References

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Author Note

This practitioner case study is derived from the author's MSc Cybersecurity project at the University of Hertfordshire. It is intended to communicate the methodology and findings to practitioners and employers and is not a substitute for the full academic dissertation, legal advice, or validated operational forensic guidance.